

A COUNTEREXAMPLE TO A CONVEXITY QUESTION OF BŁOCKI AND ZWONEK

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ABSTRACT. In [4], Błocki and Zwonek raised the question whether, for a pseudoconvex domain $D \subset \mathbb{C}^n$ containing 0, a homogeneous polynomial H of degree k , and the scaled sublevel sets $D_a := e^{-a}\{G_D(\cdot, 0) < a\}$, $a \leq 0$, of the pluricomplex Green function with pole at 0, the function

$$(-\infty, 0] \ni a \mapsto \log K_{D_a}^H(0)$$

is convex, as it is in the case $H \equiv 1$. In this note we answer this question in the negative in dimension one. For $Q > 1$ sufficiently large we consider the planar domain

$$D_c := \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}, \quad c = 2Q,$$

whose Green function is explicit, and we show that for $w = 0$ and $H(z) = z^6$ the function $a \mapsto \log K_{(D_c)_a}^{z^6}(0)$ violates the midpoint convexity inequality. The proof combines an exact reduction of the extremal problem to a weighted Bergman kernel on the unit disk with a rigorous asymptotic expansion of the corresponding infinite Gram matrix.

1. INTRODUCTION

Let $D \subset \mathbb{C}^n$ be a bounded pseudoconvex domain and let $w \in D$. The *Bergman kernel* of D at w is defined by

$$K_D(w) := \sup \left\{ |f(w)|^2 : f \in \mathcal{O}(D), \|f\|_D := \left(\int_D |f|^2 d\lambda \right)^{1/2} \leq 1 \right\},$$

where $d\lambda$ denotes the Lebesgue measure. Following [4], for a homogeneous polynomial H of degree k , $H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$, we write $P_H(f) := \sum_{|\alpha|=k} a_\alpha D^\alpha f$, and we define the higher order Bergman kernel

$$K_D^H(z) := \sup \left\{ |P_H(f)(z)|^2 : f \in \mathcal{O}(D), f^{(j)}(z) = 0 \ (0 \leq j < k), \|f\|_D \leq 1 \right\}.$$

In dimension one, $H(z) = z^k$ and $K_D^{z^k}(z)$ will also be denoted by $K_D^{(k)}(z)$.

Let $G_D(\cdot, w)$ be the pluricomplex Green function of D with pole at w , and assume $w = 0$. For $a \leq 0$ we denote

$$D_a := e^{-a}\{G_D(\cdot, 0) < a\}.$$

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The counterexample presented in this paper was originally produced by GPT, a large language model. The author has independently verified the statement and all computations.

The main theorem of [4] asserts that $a \mapsto K_{D_a}^H(0)$ is non-decreasing, and it is noted that

$$K_{D_a}^H(0) = e^{2(n+k)a} K_{\{G_D(\cdot, 0) < a\}}^H(0).$$

In [4, Remark 3] the following question is raised:

Question 1.1 (Błocki–Zwonek, [4, Remark 3]). *Is the function*

$$(-\infty, 0] \ni a \mapsto \log K_{D_a}^H(w)$$

convex, as it is in the case $H \equiv 1$?

The convexity in the case $H \equiv 1$ follows from Berndtsson’s theorem on the plurisubharmonic variation of Bergman kernels (see the final remark in [2]). The purpose of this note is to show that Question 1.1 has a negative answer in general.

Theorem 1.2. *For $Q > 1$ sufficiently large set $c := 2Q$ and*

$$D_c := \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}.$$

Then D_c is a domain in $\mathbb{C}$ (indeed $\mathbb{C}$ minus two disjoint Jordan regions, hence unbounded and doubly connected), $0 \in D_c$, and with $H(z) = z^6$ the function

$$(-\infty, 0] \ni a \mapsto \log K_{(D_c)_a}^{z^6}(0)$$

is not convex. More precisely, for

$$a_- := -\log 2, \quad a_0 := -\frac{1}{2} \log 2, \quad a_+ := 0,$$

so that $a_0 = (a_- + a_+)/2$, we have

$$\log K_{(D_c)_{a_0}}^{z^6}(0) > \frac{1}{2} \left(\log K_{(D_c)_{a_-}}^{z^6}(0) + \log K_{(D_c)_{a_+}}^{z^6}(0) \right),$$

and the difference is

$$\frac{3}{64Q^2} + O(Q^{-4}) \quad (Q \rightarrow \infty).$$

The proof of Theorem 1.2 is divided into three steps, each of independent interest:

- (1) an explicit formula for the Green function of D_c and the observation that the scaled domains $(D_c)_a$ are again of the same form (Section 2);
- (2) an exact reduction of the extremal problem defining $K_{D_q}^{z^6}(0)$ to the third Taylor coefficient functional on a weighted Bergman space of the unit disk (Section 3);
- (3) a rigorous asymptotic expansion of the associated infinite Gram matrix, valid up to order q^{-4} , which yields the precise size of the convexity defect (Section 4).

The counterexample presented in this note was originally produced by GPT, a large language model. The author has independently verified the algebraic identities and the asymptotic expansion.

Remark 1.3 (A different convexity question). *We recall that Błocki and Zwonek also conjectured in [3] the convexity of the area function $t \mapsto \log \lambda(\{G_{\Omega, w} < t\})$; counterexamples on annuli were found numerically in [1] and proved in full generality in [3]. That conjecture concerns the volume of sublevel sets and is unrelated to Question 1.1, which concerns the Bergman kernel.*

2. THE GREEN FUNCTION OF D_c AND THE SCALING LAW

Throughout the paper we write $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$.

Proposition 2.1. *Let $c > 0$ and*

$$F_c(z) := \frac{z^2}{1 - cz^2}.$$

Then the domain $D_c = \{|F_c| < 1\}$ satisfies $0 \in D_c$, and the mapping

$$F_c : D_c \longrightarrow \mathbb{D} \setminus \{-1/c\}$$

is proper of degree 2, vanishing to order 2 at 0. Consequently the (negative) pluri-complex Green function of D_c with pole at 0 is

$$G_{D_c}(z, 0) = \frac{1}{2} \log |F_c(z)| = \log |z| - \frac{1}{2} \log |1 - cz^2|.$$

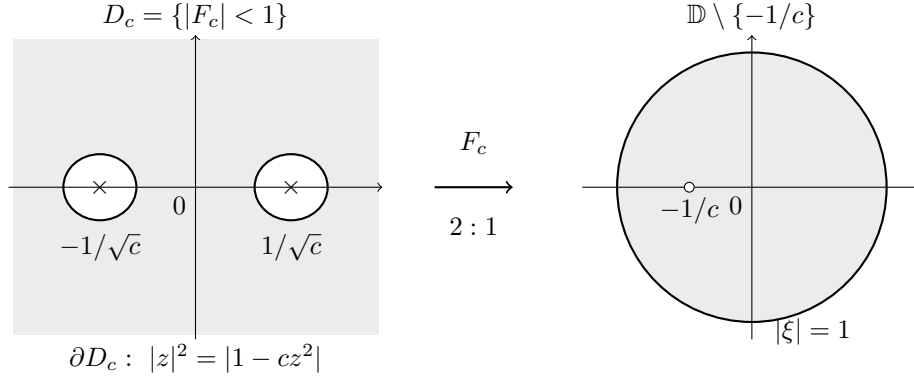


FIGURE 1. The domain $D_c = \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}$ in the z -plane (shaded; the two holes are centered at $\pm 1/\sqrt{c}$ and the boundary is $|z|^2 = |1 - cz^2|$) and its image $F_c(D_c) = \mathbb{D} \setminus \{-1/c\}$ in the ξ -plane. The map F_c is proper of degree 2 with $F_c(0) = 0$ (a double zero) and $F_c(\infty) = -1/c$.

Proof. The identity $D_c = \{|F_c| < 1\}$ is immediate from the definitions. The rational function F_c is holomorphic on D_c : the poles $\pm c^{-1/2}$ are not contained in D_c , since $|F_c| \rightarrow +\infty$ near them. The equation $F_c(z) = \xi$ reads $z^2 = \xi(1 - cz^2)$, i.e. $z^2 = \xi/(1 + c\xi)$, which has two solutions for every $\xi \notin \{-1/c, \infty\}$, so F_c is proper of degree 2 and $F_c(0) = 0$ with $F'_c(0) = 0$, $F''_c(0) = 2 \neq 0$.

For a proper holomorphic map of degree m between planar domains with $f(0) = 0$, the Green functions satisfy $G_D(z, 0) = \frac{1}{m} G_{D'}(f(z), 0)$. Since $\mathbb{D} \setminus \{-1/c\}$ is obtained from $\mathbb{D}$ by removing a polar set, its Green function with pole 0 is $\log |\cdot|$. Taking $m = 2$ gives the formula. $\square$

For $a \leq 0$ we use the notation of [4]:

$$(D_c)_a := e^{-a} \{G_{D_c}(\cdot, 0) < a\}.$$

Proposition 2.2. *For $a \leq 0$ and $q := ce^{2a}$ we have*

$$(D_c)_a = D_q := \{x \in \mathbb{C} : |x|^2 < |1 - qx^2|\}.$$

Proof. By Proposition 2.1, $G_{D_c}(z, 0) < a$ is equivalent to $|F_c(z)| < e^{2a}$. Writing $z = e^a x$, we have

$$F_c(e^a x) = e^{2a} \frac{x^2}{1 - ce^{2a} x^2} = e^{2a} F_q(x),$$

with $q = ce^{2a}$. Hence $|F_c(e^a x)| < e^{2a}$ is equivalent to $|F_q(x)| < 1$, i.e. $x \in D_q$. Since $(D_c)_a = \{x : e^a x \in \{G_{D_c}(\cdot, 0) < a\}\}$, the claim follows. $\square$

Since $\log q = \log c + 2a$ is affine in a , the convexity of $a \mapsto \log K_{(D_c)_a}^{z^6}(0)$ is equivalent to the convexity of $s \mapsto \log K_{D_{e^s}}^{z^6}(0)$ in $s = \log q$. We shall therefore study the family D_q , $q > 1$.

3. REDUCTION TO A WEIGHTED BERGMAN KERNEL ON THE UNIT DISK

For $q > 1$ we abbreviate $K_q := K_{D_q}^{z^6}(0)$. Since D_q is invariant under $x \mapsto -x$, the sixth derivative at 0 only sees the even part of a holomorphic function, and we may restrict to

$$f(x) = g(x^2), \quad g \in \mathcal{O}(U_q),$$

where

$$U_q := \{y \in \mathbb{C} : |y| < |1 - qy|\} = \mathbb{C} \setminus \overline{B\left(\frac{q}{q^2 - 1}, \frac{1}{q^2 - 1}\right)}$$

is the image of D_q under the two-to-one map $y = x^2$.

We record the elementary facts used below.

Lemma 3.1. *With the notation above, the following hold.*

(i) *For $f(x) = g(x^2)$,*

$$\|f\|_{L^2(D_q)}^2 = \frac{1}{2} \int_{U_q} |g(y)|^2 \frac{dA(y)}{|y|}.$$

(ii) *The conditions $f(0) = f'(0) = \dots = f^{(5)}(0) = 0$ and $f^{(6)}(0) = 1$ are equivalent to*

$$g(y) = b_3 y^3 + O(y^4) \quad (y \rightarrow 0), \quad b_3 = \frac{1}{6!}.$$

(iii) *Under the Möbius transformation*

$$\zeta = \frac{y}{1 - qy}, \quad y = \frac{\zeta}{1 + q\zeta},$$

which maps U_q biholomorphically onto $\mathbb{D} \setminus \{-1/q\}$, the square integrability of $|g|^2/|y|$ forces $g(y(\zeta))$ to vanish at $\zeta = -1/q$. Hence there exists $h \in \mathcal{O}(\mathbb{D})$ with

$$g(y(\zeta)) = (1 + q\zeta)h(\zeta),$$

and

$$\|f\|_{L^2(D_q)}^2 = \frac{1}{2} \int_{\mathbb{D}} |h(\zeta)|^2 \frac{dA(\zeta)}{|\zeta| |1 + q\zeta|}.$$

(iv) *In the variable ζ , the normalization $b_3 = 1/6!$ becomes*

$$h(\zeta) = \frac{1}{6!} \zeta^3 + O(\zeta^4).$$

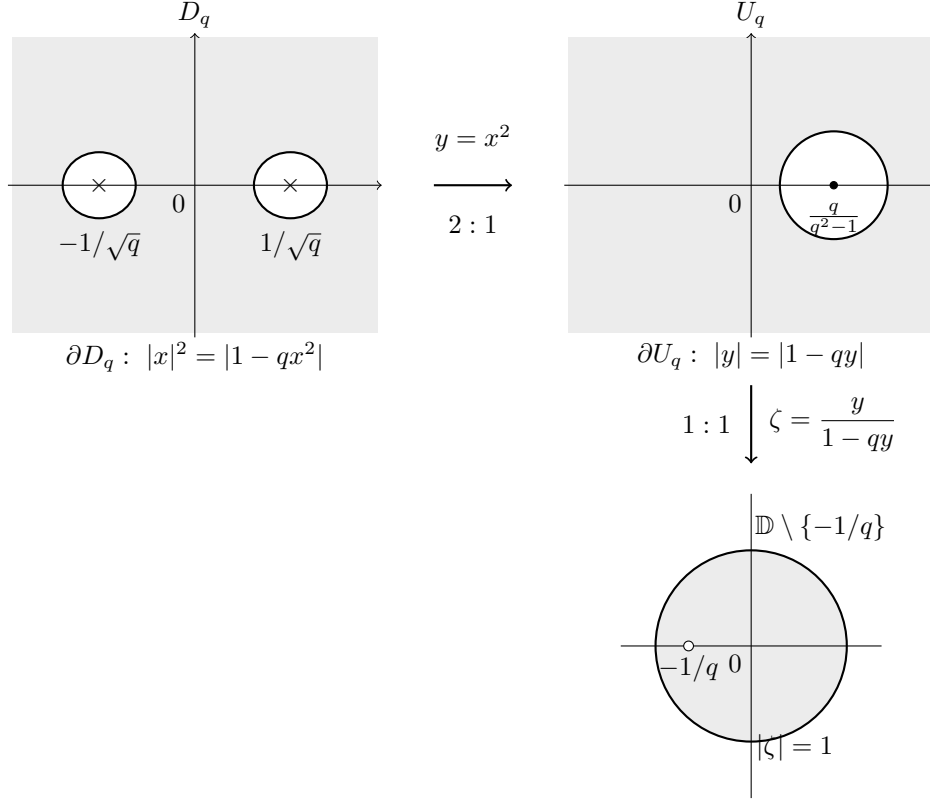


FIGURE 2. The domain $D_q = \{x \in \mathbb{C} : |x|^2 < |1 - qx^2|\}$ in the x -plane (shaded; the two holes are centered at $\pm \sqrt{q/(q^2 - 1)}$ and contain the poles $\pm 1/\sqrt{q}$), its image $U_q = \{y \in \mathbb{C} : |y| < |1 - qy|\} = \mathbb{C} \setminus \overline{B(q/(q^2 - 1), 1/(q^2 - 1))}$ in the y -plane under the two-to-one map $y = x^2$, and the unit disk $\mathbb{D} \setminus \{-1/q\}$ in the ζ -plane obtained via the Möbius map $\zeta = y/(1 - qy)$ (1 : 1).

Proof. (i) The map $y = x^2$ is two-to-one from D_q onto U_q , and $dA(y) = 4|x|^2 dA(x) = 4|y| dA(x)$, whence $dA(x) = dA(y)/(4|y|)$.

(ii) Writing $g(y) = \sum_{m \geq 0} g_m y^m$, we have $f(x) = \sum_m g_m x^{2m}$; the first six derivatives vanish at 0 iff $g_0 = g_1 = g_2 = 0$, and $f^{(6)}(0) = 6! g_3$.

(iii) The Möbius map is a biholomorphism between U_q and $\mathbb{D} \setminus \{-1/q\}$ because $y = \zeta/(1 + q\zeta)$ and

$$1 - qy = \frac{1}{1 + q\zeta}, \quad |y| < |1 - qy| \iff |\zeta| < 1.$$

Near $\zeta = -1/q$ one has $y \sim -1/(q^2(\zeta + 1/q)) \rightarrow \infty$. Since g is holomorphic on U_q , including the point at infinity, $g(y(\zeta))$ is holomorphic near $\zeta = -1/q$; if it did not vanish there, then $|g|^2/|y| \sim |g(-1/q)|^2 |1 + q\zeta|^3$ would not be integrable near $\zeta = -1/q$, contradicting the finiteness of $\|f\|^2$. Hence $g(y(\zeta))$ vanishes at $\zeta = -1/q$, so the quotient $h = g(y(\zeta))/(1 + q\zeta)$ is holomorphic on $\mathbb{D}$. The change of variables

$y = \zeta/(1 + q\zeta)$ has Jacobian $dA(y) = |1 + q\zeta|^{-4} dA(\zeta)$, and $|y|^{-1} = |1 + q\zeta|/|\zeta|$, so

$$\frac{1}{2} \int_{U_q} |g|^2 \frac{dA(y)}{|y|} = \frac{1}{2} \int_{\mathbb{D}} |h|^2 \frac{dA(\zeta)}{|\zeta| |1 + q\zeta|}.$$

(iv) From $y = \zeta + O(\zeta^2)$ we get $g(y(\zeta)) = g_3 \zeta^3 + O(\zeta^4)$; dividing by $(1 + q\zeta)$ does not change the leading term. $\square$

We now introduce the infinite Gram matrix. For $i, j \geq 3$ set

$$M_{ij}(q) := \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta| |1 + q\zeta|} dA(\zeta),$$

and let $K_3(q) := (M(q)^{-1})_{33}$ be the $(3, 3)$ -entry of the inverse of the infinite matrix $M(q)$ (which is positive definite on the space of holomorphic functions h with $\int |h|^2 / (|\zeta| |1 + q\zeta|) dA < \infty$).

Proposition 3.2. *For every $q > 1$,*

$$K_q = K_{D_q}^{z^6}(0) = 2(6!)^2 K_3(q).$$

Proof. By definition,

$$K_q^{-1} = \min \left\{ \|f\|_{L^2(D_q)}^2 : f \in \mathcal{O}(D_q), f^{(j)}(0) = 0 \ (0 \leq j < 6), f^{(6)}(0) = 1 \right\}.$$

By Lemma 3.1 this equals

$$\min \left\{ \frac{1}{2} \int_{\mathbb{D}} |h|^2 \frac{dA}{|\zeta| |1 + q\zeta|} : h \in \mathcal{O}(\mathbb{D}), h(\zeta) = \frac{1}{6!} \zeta^3 + O(\zeta^4) \right\}.$$

Writing $h(\zeta) = \frac{1}{6!} \zeta^3 + \sum_{m \geq 4} c_m \zeta^m$, the minimum over the higher coefficients is

$$\frac{1}{2(6!)^2} (M(q)^{-1})_{33},$$

and taking the reciprocal gives the claim. $\square$

4. ASYMPTOTICS OF THE GRAM MATRIX

Set $\varepsilon := 1/q$ and $A(\varepsilon) := qM(q)$, so that

$$A_{ij}(\varepsilon) = \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta| |\zeta + \varepsilon|} dA(\zeta), \quad i, j \geq 3.$$

We write $D = \text{diag}(\pi/3, \pi/4, \pi/5, \dots)$, and for the off-diagonal terms we use the convention that $B_{i,i+1} = B_{i+1,i}$ for $i \geq 3$.

Proposition 4.1. *The matrix function $A(\varepsilon)$ admits the expansion*

$$A(\varepsilon) = D + \varepsilon B + \varepsilon^2 C + \varepsilon^3 E + O(\varepsilon^4)$$

in the sense of bounded quadratic forms on the Hilbert space $\{h = \zeta^3 v : v \in L_h^2(\mathbb{D})\}$ with norm $\int |h|^2 |\zeta|^{-2} dA$. The nonzero leading terms are

$$D_{ii} = \frac{\pi}{i}, \quad B_{i,i+1} = B_{i+1,i} = -\frac{\pi}{2i}, \quad C_{ii} = \frac{\pi}{4(i-1)}, \quad i \geq 3.$$

Consequently, for $\varepsilon > 0$ sufficiently small,

$$(A(\varepsilon)^{-1})_{33} = \frac{3}{\pi} - \frac{1}{8\pi} \varepsilon^2 + O(\varepsilon^4).$$

Proof. The expansion of the kernel in the outer region $|\zeta| \geq 2\varepsilon$,

$$\frac{1}{|\zeta + \varepsilon|} = \frac{1}{|\zeta|} \left(1 + \frac{2\varepsilon \operatorname{Re} \zeta}{|\zeta|^2} + \frac{\varepsilon^2}{|\zeta|^2} \right)^{-1/2} = \frac{1}{|\zeta|} - \frac{\varepsilon \operatorname{Re} \zeta}{|\zeta|^3} + \varepsilon^2 \frac{3(\operatorname{Re} \zeta)^2 - |\zeta|^2}{2|\zeta|^5} + O\left(\frac{\varepsilon^3}{|\zeta|^4}\right),$$

together with the elementary bound in the inner disk $|\zeta| < 2\varepsilon$ (where $\zeta = \varepsilon s$, so that all quadratic forms under consideration are $O(\varepsilon^6)$ after testing against $h = \zeta^3 v$), justifies the expansion in the sense of bounded quadratic forms; the remainder is controlled by

$$C\varepsilon^4 \int_{\mathbb{D}} |v|^2 dA \leq C'\varepsilon^4 \int_{\mathbb{D}} |h|^2 |\zeta|^{-2} dA.$$

The angular Fourier integrals give

$$\int_0^{2\pi} e^{id\theta} d\theta = 2\pi\delta_{d0}, \quad \int_0^{2\pi} \cos \theta e^{i\theta} d\theta = \pi, \quad \int_0^{2\pi} (3\cos^2 \theta - 1) d\theta = \pi,$$

which yield, after the radial integration,

$$\begin{aligned} D_{ii} &= \int_0^1 \rho^{2i} \frac{2\pi}{\rho} d\rho = \frac{\pi}{i}, \\ B_{i,i+1} &= - \int_0^1 \rho^{2i+1} \frac{\pi}{\rho^2} d\rho = -\frac{\pi}{2i}, \\ C_{ii} &= \int_0^1 \rho^{2i} \frac{\pi}{2\rho^3} d\rho = \frac{\pi}{4(i-1)}. \end{aligned}$$

The first-order off-diagonal entries of B with $|i-j| \neq 1$ vanish by angular orthogonality.

For the inverse, we use the Neumann expansion $A^{-1} = D^{-1} - \varepsilon D^{-1} B D^{-1} + \varepsilon^2 (D^{-1} B D^{-1} B D^{-1} - D^{-1} C D^{-1}) + O(\varepsilon^3)$. Since D is diagonal, the second-order contribution to the $(3,3)$ -entry receives only the diagonal term C_{33} and the unique off-diagonal path $3 \rightarrow 4 \rightarrow 3$, so that

$$\begin{aligned} (A^{-1})_{33} &= \frac{3}{\pi} + \varepsilon^2 \left(\frac{B_{34}^2}{D_{33}^2 D_{44}} - \frac{C_{33}}{D_{33}^2} \right) + O(\varepsilon^4) \\ &= \frac{3}{\pi} + \varepsilon^2 \left(\frac{(\pi/6)^2}{(\pi/3)^2 (\pi/4)} - \frac{\pi/8}{(\pi/3)^2} \right) + O(\varepsilon^4) \\ &= \frac{3}{\pi} - \frac{1}{8\pi} \varepsilon^2 + O(\varepsilon^4). \end{aligned}$$

There is no third-order term: the transformation $\zeta \mapsto -\zeta$ unitarily identifies $A(\varepsilon)$ with $A(-\varepsilon)$, while the coefficient functional is unchanged, so the expansion of $(A^{-1})_{33}$ contains only even powers. $\square$

Combining Proposition 3.2 with Proposition 4.1 gives the asymptotic law for the kernel.

Corollary 4.2. *As $q \rightarrow \infty$,*

$$K_3(q) = q(A(q^{-1})^{-1})_{33} = \frac{3q}{\pi} \left(1 - \frac{1}{24q^2} + O(q^{-4}) \right),$$

and therefore, with $C_0 := \log(2(6!)^2 \cdot 3/\pi)$,

$$\log K_{D_q}^{\mathbb{Z}_6}(0) = C_0 + \log q - \frac{1}{24q^2} + O(q^{-4}).$$

5. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.2. Let $Q > 1$ and $c = 2Q$. By Proposition 2.2, the three values

$$a_- = -\log 2, \quad a_0 = -\frac{1}{2} \log 2, \quad a_+ = 0$$

correspond to

$$q_- = \frac{Q}{2}, \quad q_0 = Q, \quad q_+ = 2Q.$$

Since $a_0 = (a_- + a_+)/2$ and $\log q = \log c + 2a$ is affine in a , it suffices to prove

$$\log K_{D_Q}^{z^6}(0) - \frac{1}{2} \left(\log K_{D_{Q/2}}^{z^6}(0) + \log K_{D_{2Q}}^{z^6}(0) \right) > 0$$

for Q sufficiently large. By Corollary 4.2, the terms C_0 cancel, and so do the linear terms $\log q$:

$$\log Q - \frac{1}{2} (\log(Q/2) + \log(2Q)) = 0.$$

The remaining contribution is

$$-\frac{1}{24} \left(\frac{1}{Q^2} - \frac{1}{2} \left(\frac{4}{Q^2} + \frac{1}{4Q^2} \right) \right) + O(Q^{-4}) = \frac{3}{64Q^2} + O(Q^{-4}) > 0$$

for Q large enough. This proves the strict midpoint inequality, hence the failure of convexity. $\square$

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